

Research Project: Franciscan Eco-Theology and Environmental Attitudes

A term paper submitted for the
RM in Management and Applied Microeconomics

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Abstract

Our project focuses on the effect of exposure to the main Mendicant orders of the 12th and 15th century - the Franciscan and Dominican - on the contemporary environmental attitudes. We find that that higher Franciscan is associated with less pro-environmental attitudes, while higher Dominican exposure is associated with more pro-environmental attitudes.

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1 Introduction

2 Context and Hypotheses

We will first explore the historical context of the two main Mendicant orders - the Franciscan and the Dominican - to formulate our hypothesis on how those theological beliefs influenced the environmental attitudes.

2.1 The Franciscan Order

The Franciscan Order was founded in the early 13th century by St. Francis of Assisi (1181-1226), who was recognized by the Catholic Church as the patron saint of ecology, widely known for his profound affection towards nature and animals. In his "Canticles of the Creatures," St. Francis refers to the sun and the moon as brother and sister. Thus, the Franciscan Order emphasized the imitation of Christ through the ideals of poverty and humility. They focus on collective life, communal care, antimaterialism, shame, compassion, simplicity, and humbleness (Arrunada and Lopez-Manuel, 2024).

2.2 The Dominican Order

In contrast, the Dominican Order, founded by St. Dominic (1170-1221) in the early 13th century, prioritized intellectual and educational mission of medieval Catholic Church. They founded and staffed universities and produced extensive theological and practical writings (Le Goff, 1980; Boyle 1981). The Dominicans emphasized the importance of critical thinking, self-examination, and moral development (Hinnebusch 1966; Lesnick 1989).

2.3 Hypothesis

Because of the strong association of the Franciscan Order with nature and animals, we would naturally expect more-environmental attitudes in regions with stronger Franciscan exposure. Therefore, we initially hypothesized that higher historical exposure to the Franciscan Order would be associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order would be associated with less pro-environmental attitudes. However, our results state the opposite.

3 Literature Review

A similar study was conducted by Arruñada and López-Manuel (2024), who examined the effects of the two main Mendicant orders of the Late Middle Ages - the Dominican and the Franciscan. Through their emphasis on guilt, shame, compassion, and analytical thinking, these orders shape outcomes at the cognitive, interpersonal, and institutional levels, which together constitute the “foundations” of impersonal exchange. They further argue that eco-theological beliefs can foster distinct values and cultural norms. For example, at the interpersonal level, “Dominicans promoted guilt as a driver of prosocial behavior toward strangers”, fostering a culture of individual accountability and independence, “whereas Franciscans encouraged compassion and shame, which strengthened bonds within the group” (Arruñada and López-Manuel, 2024). Consistent with these differences, the authors document contrasting effects of Dominican and Franciscan influence on impersonal exchange. Thus, their findings supports our results for distinct effects of the two orders on environmental attitudes.

Our study also relates to the broader literature on the role of Medieval Churches and persistence of cultural norms. The Church, as an institution, played a pervasive role in every aspect of life in medieval Europe, shaping people’s lives, values, and behavior throughout the Middle Ages (Lynch, 2024). Once a religious institution becomes established, the values and norms it historically imposes tend to persist, as such cultural legacies “leave an imprint on values that endures” over time (Inglehart and Baker, 2000). Although religious institutions have lost some of the formal allegiance of their followers in modern societies, Inglehart and Baker’s findings suggest that religiously rooted value orientations continue to influence individuals, as these values become embedded in national cultures and are transmitted through educational institutions and mass media, allowing religiously rooted value orientations to persist at the individual level.

Another important strand of literature relates to the determinants of environmental attitudes. In many literatures it was examined that cultural differences might influence environmental attitudes (Chwialkowska et al., 2020; Khan et al., 2024; Yang et al., 2024; Tam, 2025). Some theoretical models argue that pro-environmental attitudes may be rooted in self-interest and utility maximization. In particular, a longer life expectancy may increase individuals’ willingness to support environmental efforts to fight climate change (Mariani et al., 2010). Additionally, importantly for our research, many other literature with similar research interests commonly measure the environmental attitudes using the survey datasets (EVS) (Gelissen, 2007; Behler, 2025).

Our research is the first to examine the effect of exposure to the two main Mendicant orders of the Medieval Ages - the Franciscan and the Dominican - on contemporary environmental attitudes.

4 Data and Methodology

4.1 Exposure Measure

In our research, we assume that people that are living close to the monasteries can get influenced or exposed to the theological beliefs of the monasteries. Thus, the exposure measure is defined as a fraction of a region's historical population living geographically close to a monastic house. To estimate historical exposure to the Franciscan and Dominican orders, we borrow from a lot of datasets. First, we employ the geodataset covering the years between 1216 to 1500 from the Mapping Past Societies (MAPS), which includes information on the geographic locations (latitude and longitude) of the Dominican (870 observations) and Franciscan houses (994 observations) and the name of the houses. The Dominican dataset includes the foundation centuries. The only limitation of this dataset is that it does not include the ending years. After we get the monastery location, we need to locate them within regions. To do so, we use the European Commission system for characterising geographical data - Nomenclature of Territorial Units for Statistics (NUTS), as well as country shapefile. NUTS 1 is for large macro regions, like states, NUTS 2 is for more localized regions within for all of the countries. Since Germany has modified their administrative borders, we rely only on the larger one - NUTS 1 regions. We needed to delete monasteries in east Ukraine, Crimea, and Middle East cities (Jerusalem/Acre), since those monasteries lie in the locations that are not in the European Union. There are some monasteries that are a little bit outside the EU; we coerce them to fit in regions, such as Romania or Poland or Greece, because they're pretty close border wise.

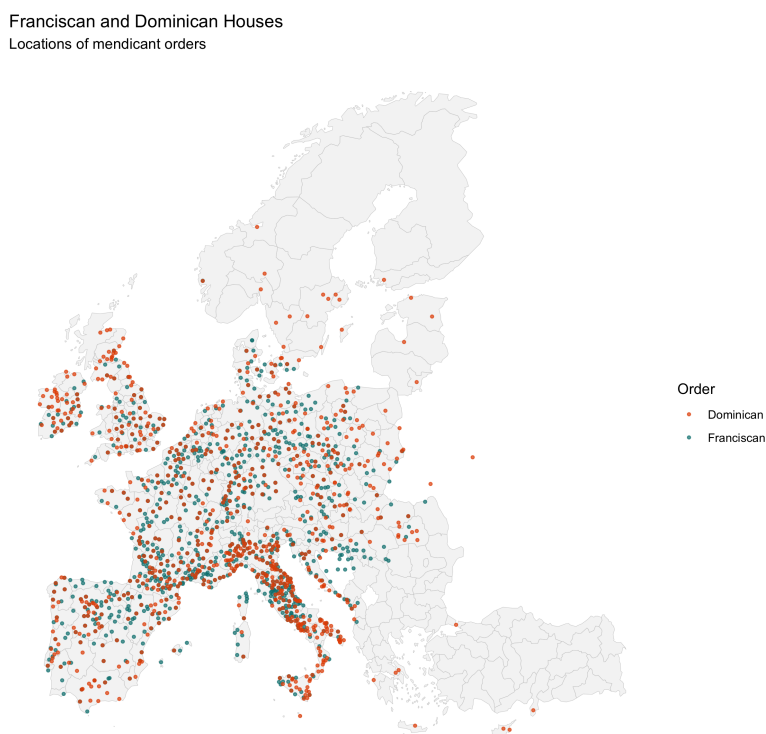


Figure 1: Map of the European Union with Franciscan and Dominican Monasteries

Once we have combined the monastery locations with NUTS and shapefile, our next step is to observe historical population data .

HYDE 3.3 (History Database of the Global Environment) Utrecht University population data captures where people lived historically. It is an estimate of population count and density on regular latitude-longitude grid from the very late Mediaval time to the modern period. Grids are very small squares of 85 km², and we have an estimate of population density for each grid.

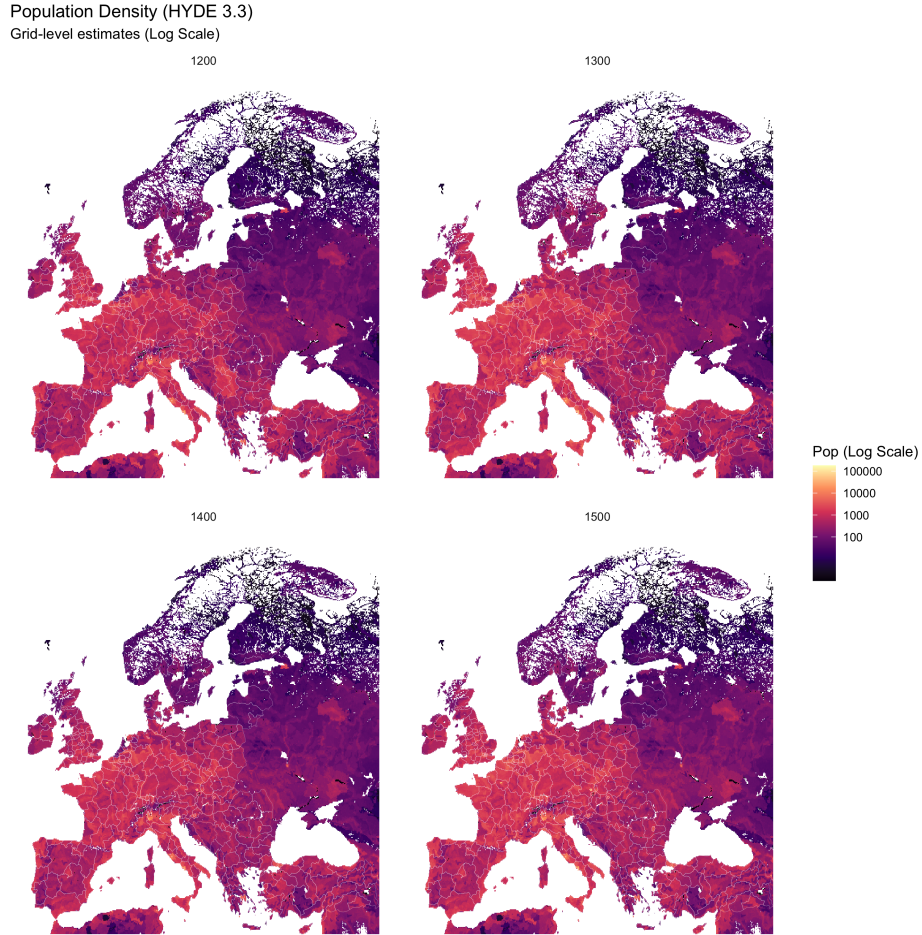


Figure 2: Map of the Population densities (Grid-level)

Similarly, we can think of it not in squares, but in the NUTS regions. By doing similar calculations, we can get the exposure measure for each NUTS 2 region (Figure 3).

We draw a circle around the monastery and all the little squares (grids) that are inside that circle are counted as exposed population. In our research, the exposure measure is defined as the sum of exposed population within a given region as a percentage of the total population for both Franciscan and Dominican monasteries:

$$\text{Exposure} = \frac{\text{Exposed Population}}{\text{Total Population}}$$

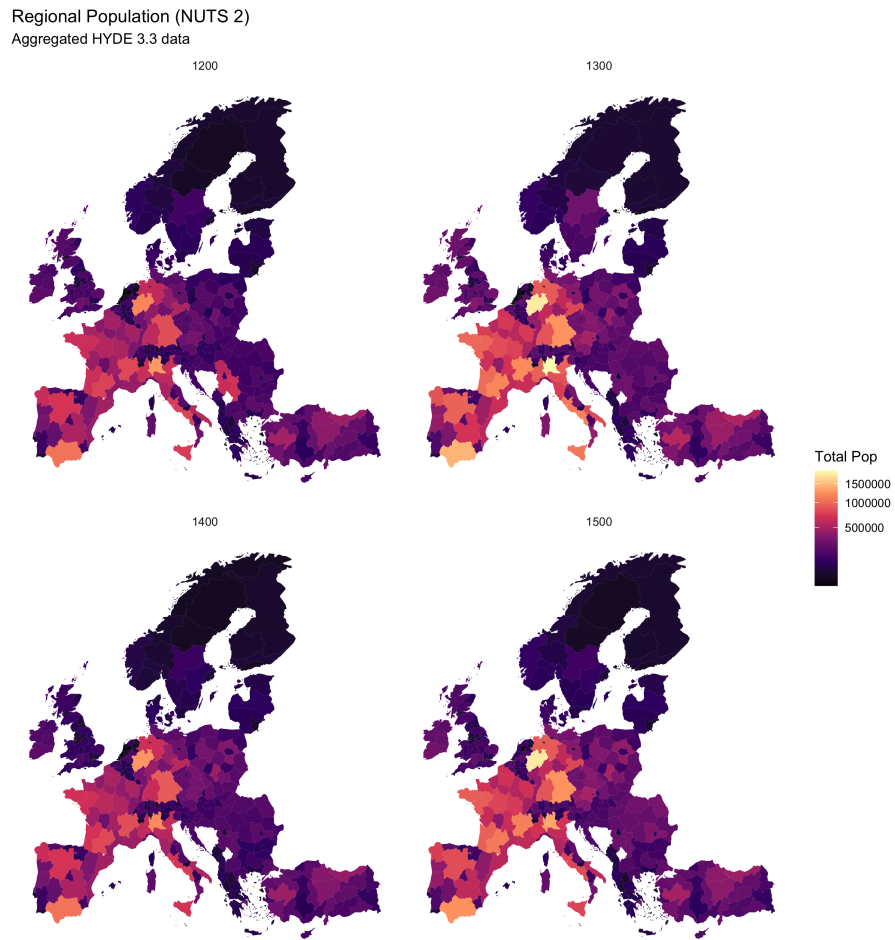


Figure 3: Map of Population Densities (NUTS 2 level)

Having all the three sources put together, we have Dominican and Franciscan exposure measure for Europe (Figures 4 and 5).

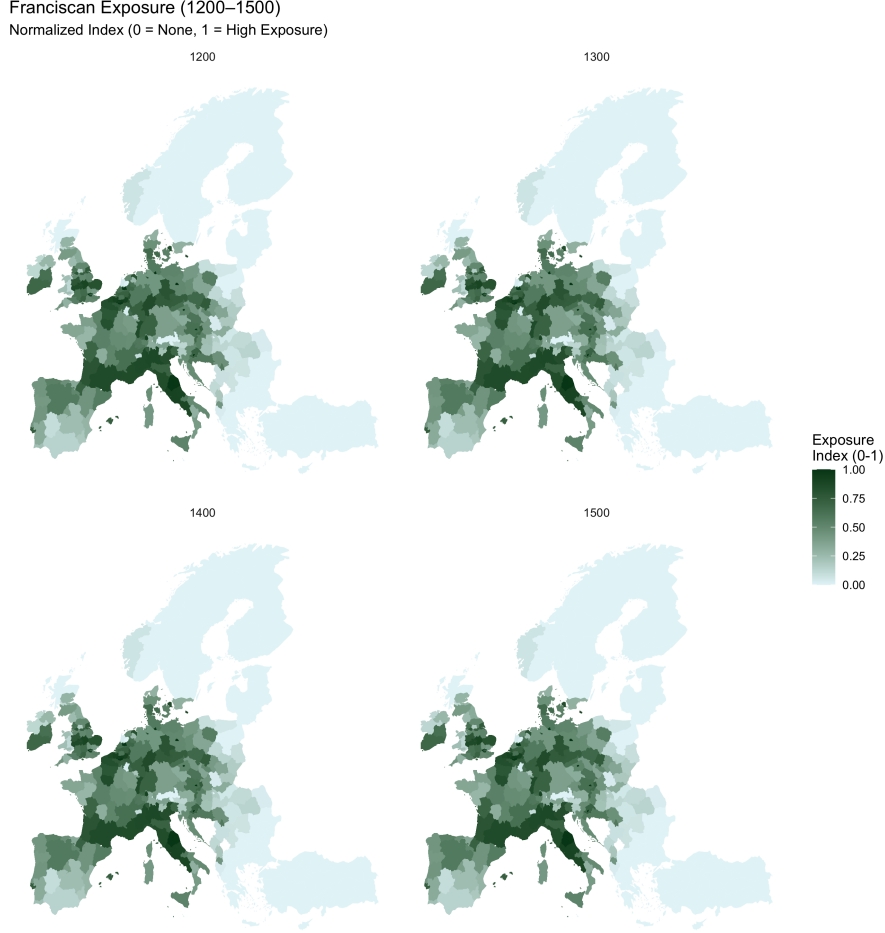


Figure 4: Map of Franciscan exposure

4.2 Environmental Attitudes Measure

To construct our main dependent variable - environmental attitudes, we use the European Values Study (EVS) dataset 2017–2021, which is individual-level survey data on values, beliefs, attitudes, and socioeconomic characteristics across European countries. The dataset covers over 30 European countries (of which 24 are relevant for our research, excluding the countries that are not in NUTS-2). After the data cleaning, the dataset includes 49233 observations, and it includes geographic identifiers for NUTS-1 (Germany) and NUTS-2 regions.

From the EVS dataset, we borrow particularly 6 environmental variables, which are the questions that are related to environmental attitudes. The questions are the following:

- Choosing either "to protect the environment, even if slower economic growth and loss of jobs" or "to have economic growth and newly created jobs, even if environment suffers".
 - Economy and Environment Trade-off question
- "I would give part of my income if I were certain that the money would be used to prevent pollution."

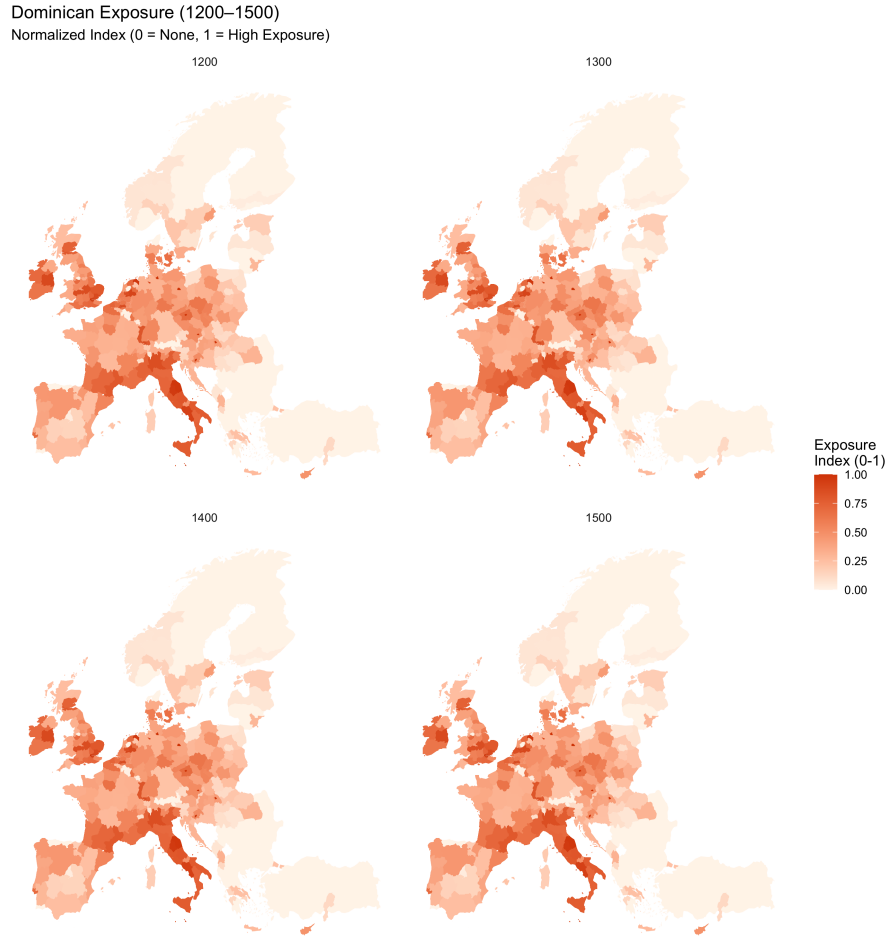


Figure 5: Map of Dominican exposure

- "It is just too difficult for someone like me to do much about the environment."
- "There are more important things to do in life than protecting the environment."
- Tells us a bit on people's preferences
- "There is no point in doing what I can for the environment unless others do the same."
- Network effect and Collective action problem question
- "Many of the claims about environmental threats are exaggerated."

The questions have categorical answers ("strongly agree", "agree", "neither agree nor disagree", "disagree", "strongly disagree", or choosing between the two statements). However, the questions differ in their directional interpretation. Thus, we need to standardize the questions by assigning 0-1 scale per answer for each question, such that the answers indicate the same environmental direction. In the standardized 0-1 scale, 1 indicates the least pro-environmental attitude, and 0 indicates the most pro-environmental attitude. See the Figure 6 for visualization for environmental questions within NUTs 2 regions. The green regions are more pro-environmental, while the yellow regions are less pro-environmental.

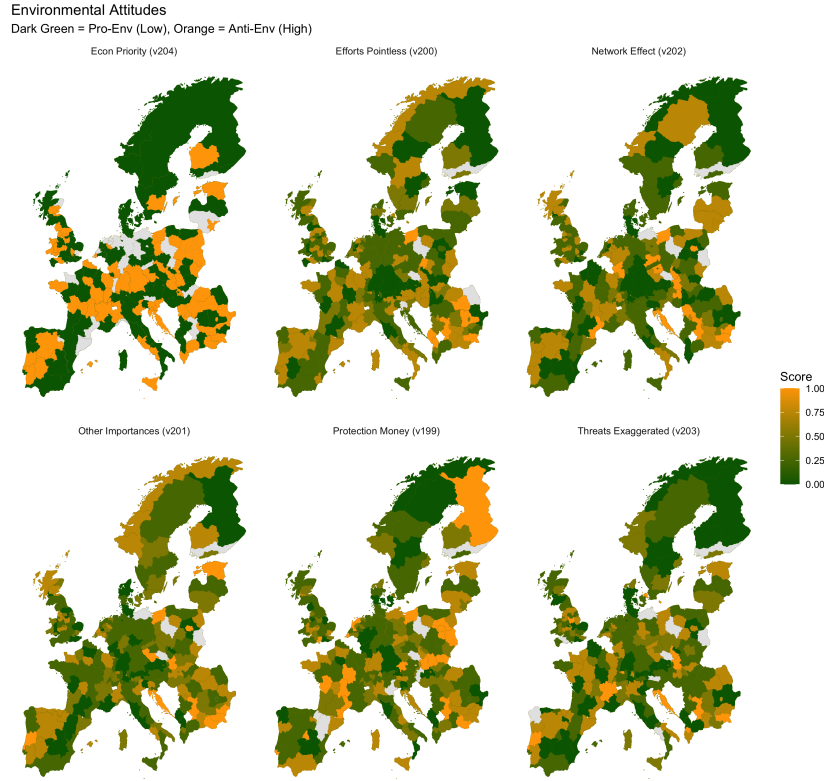


Figure 6: Distribution of the 6 environmental questions across Europe considering the average scores per region.

Since we have 6 different closely related questions for environmental attitudes, we want to reduce the dimensionality of the data and construct a single Environmental Attitudes Index. Thus, we run a Principal Component Analysis (PCA) to try to extract the main components of the questions (invisible or underlying mechanisms that could drive the answers). We find that the components of the Western European countries behaving more green, while the Eastern European countries behaving more yellowish (see the Figure 7).

Environmental Index (PCA)
Regional Averages: Dark Green = Pro-Env, Orange = Anti-Env

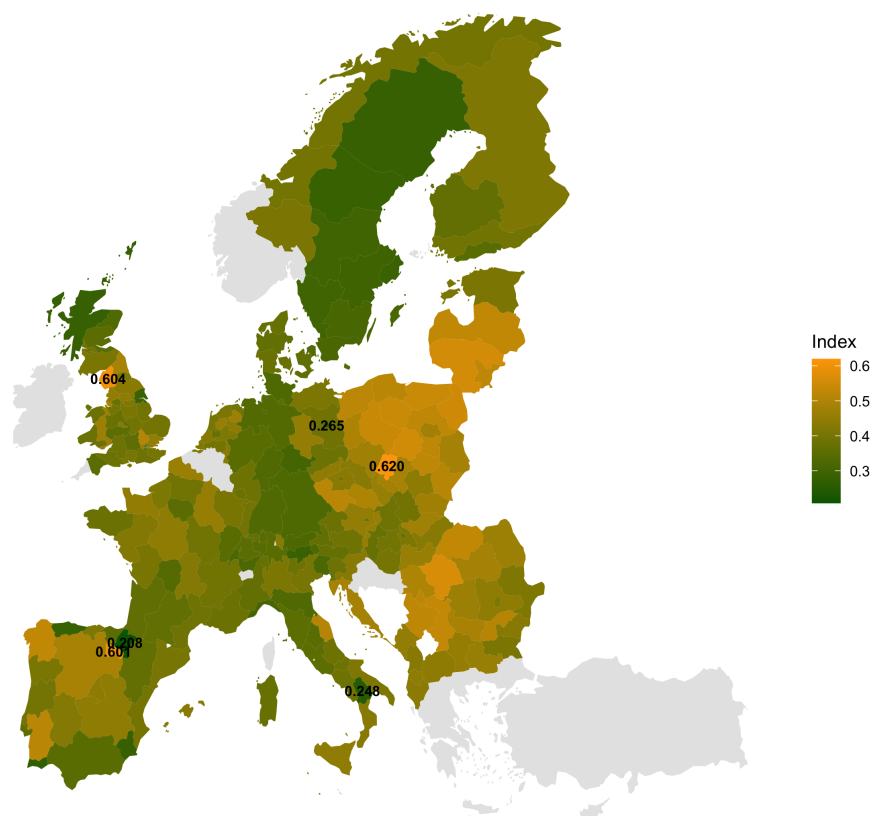


Figure 7: Environmental Index (PCA) across Europe considering the average scores per region.

5 Results

6 Limitations and Further Research

7 Conclusion

8 References